CS435 SUBJECTIVE + OBJECTIVE FILES FINALL TERM

116 - 288.

Important MCQS

1. MCQ: **MTBF** stand for_____.

Correct Answers **mean time between failure**

2. MCQ**: CSP** stand for_____.

Correct Answers **Cloud Service Provider**

3. MCQ**: SSL** stand for _____.

Correct Answers **Secure Socket Layer**

4. MCQ: **CIS** stand for_____.

Correct Answers **Cloud Information Service**

5. MCQ: The _____is called Cloudlet.

Correct Answers **workload unit**

6. MCQ : The characteristics of big data can be represented by _____.

Correct Answers **4vs**:

7. MCQ: _____consideration is applied to ensure SLA is not violated.

Correct Answers **QoS**

8. MCQ: _____ scheduling, the task starvation can be avoided.

Correct Answers: **Priority-Based**

9. MCQ: SLA violation when IT resources are_____.

Correct Answers**: overloaded**

10. MCQ: Effective resource scheduling reduces_____.

Correct Answers: **Execution time**

11. MCQ : The best option is to move the data center to _____.

Correct Answers: **Cloud**

12. MCQ: It is a third party process of finding the appropriate Cloud provider.

Correct Answers: **: Cloud Brokerage**

13. MCQ: The _____clustering creates a high-availability cluster across multiple physical servers.

Correct Answers**: Hypervisor**

14. MCQ: VMs hosted on a physical server are managed by_____.

Correct Answers**: hypervisor**

15. MCQ : Advantages of Mobile Cloud Computing:

Correct Answers**: Extending battery lifetime**

16. MCQ: _____architecture is used for provisioning of dedicated IT resources.

Correct Answers**: Resource reservation**

17. MCQ: SDN stand for_____.

Correct Answers: **Software Defined Networking**

18. MCQ: API stand for_____.

Correct Answers: **application program interface**

19. _____can be provided to an overloaded application on the recommendation and budget of the PaaS consumer.
    Correct Answers **Scalability**

20. The forensic analysis of _____is the responsibility of the consumer. Correct Answers**: IaaS**

21. The forensic analysis for _____is the responsibility of the provider. Correct Answers: **SaaS**

22. _____are usually related to quality metrics such as performance, reliability and maintainability.

Correct Answers: Non-functional requirement

23. _____are provided by the system analyst to the designer.

Correct Answers: **functional requirement**

24. VM is migrated to other server for_____.
    Correct Answers**: Load Balance**

25. WEP stand for_____.
    Correct Answers: **Wired Equivalent Privacy**

26. WPA stand for_____.
    Correct Answers ; **Wi-Fi Protected Access**

27. it may cause SLA violation when IT resources are _____.

Correct Answers: **overloaded**

28. initial investment regarding IT resource acquiring and installations are Called ?

Correct Answers: **Upfront Costs**

29. STEM stand for_____.

Correct Answers: Science, Technology, Engineering and Mathematics

30. CloudSim requires Sun's_____.

Correct Answers: Java 8 or newer version

31. _____architecture is used for provisioning of dedicated IT resources. Correct Answers: Resource reservation

32. The _____scheduling can achieve low cost and meet deadline if negotiation is successful.
    Correct Answers: bargain-based

33. _____is the process of perfecting an algorithm to make it more beneficial. Correct Answer: Optimization

34. The Cloud computing and Mobile Cloud computing are dependent upon _____usage of IT resources offered by Cloud.
    Correct Answers: remote

35. In case of Cloud computing, the providers of IaaS and PaaS manage to provide _____uptime through server redundancy and failover systems.
    Correct Answers: 99.9%

36. Load balancer module specially in case_____ scaling.
    Correct Answers: Horizontal

37. Consumers should consider the use of advanced procedures for _____provided by some providers to avoid account hijacking or identity thefts.
    Correct Answers: authentications

38. _____resource scheduling prioritizes the processing deadline of the users' requests.
    Correct Answers: Time-Based Resource

39. Include the running costs of the IT resources e.g., licensing fee, electricity, insurance and labor.

Correct Answer: On-going Costs

40. Up to _____of Europeans.
Correct Answer: 70%

41. _____excludes the access privileges of terminated employees as quickly as possible to prevent any damages.

Correct Answer: Identity as a Service (IDaaS)

42. _____is a possibility of harm or loss as a result of an activity.
Correct Answer: Risk

43. Cloud not only provides the facility of data access over the Internet.
Correct Answer: RIDA

44. The _____is configured to use all the redundant NIC. Correct Answers: virtual switch **Big data and Characteristics :**

Big Data can be: Big data is a set of techniques and technologies that require new forms of integration to uncover large hidden values from large datasets that are diverse, complex, and of a massive scale.

The following are the three generic characteristics of big data:

- Data volume is huge
- Data may not be categorized into regular relational databases ➢ The data is generated, captured and processed at high speed.

Security protocol name :

- Wired Equivalent Privacy (WEP)
- Wi-Fi Protected Access (WPA)
- Wi-Fi Protected Access 2 (WPA2)

Question: Mobile cloud computing application name?

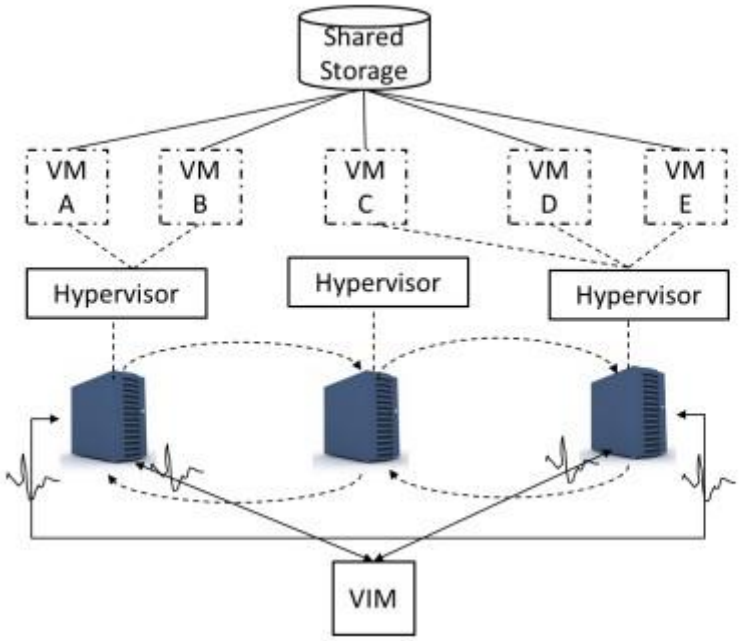
- Mobile gaming
- Mobile banking
- Mobile education
- Mobile healthcare

A few runtime configurations can be controlled by SaaS consumers. These include:

- Usage cost control
- SLA monitoring
- Security related configurations

- Occurrence Probability and Business continuity impact:

Risk	Occurrence Probability	Business Continuity Impact
User disk failure	Medium	Low
Server disk failure	Low	High
Network failure	Low	High
Database failure	Medium	High
Server power failure	High	High
Fire	Low	High
Flood	Low	High



Write the name of BOX :

1._____

2._____

3._____

4._____

Question: Accomplishment of using speed communication links for connecting cluster it resources.

- High speed communication links are used to connect the clustered IT resources for:
  - Workload distribution
  - Task scheduling
  - Data sharing
  - System synchronization

What is Cryptography?

It is a science of providing security for information. It is a science of secret communication.

- The following are the five primary functions:
  - Privacy
  - Authentication
  - Integrity
  - Non-repudiation: A mechanism to prove the originality of the sender.
  - Exchange of crypto keys which are the strings of bits used to change the format of the data.

- Effective resource scheduling reduces:
  - Execution cost
  - Execution time
  - Energy consumption

Question: On which factors cost template used by providers depend?

- The cost templates used by the providers depend upon:
  - Market competition
  - Overhead occurred during design, deployment and operations of the service
  - Cost reduction considerations through increased sharing of IT resources

- Common types of Resource Cluster Mechanism
  - Server Cluster
  - Database Cluster
  - Large Dataset Clusters

- Uses of remote administration console
  -

- Capacity planning
- Provision and releasing IT resources for on-demand usage
- Configure and setting cloud services
- Managing user accounts, security credentials, authorization and access control
- Monitor cloud service status, usage and performance

- Lack of interoperability in cloud computing (Chat GPT)
  Vendor Lock-in
  - Incompatible APIs
  - Data format in-compatibility

- VM migration across different sources:
  - Consolidation
  - Upgrade/Downgrade
  - Cloud migration
  - Disaster recovery
  - Load balancing
  - Cost Optimization

- A Cloud service may become disruptive/unavailable/down due to:
  - Over burden of processing load
  - Scheduled updates

- Cloud Balancing Architecture improves/increases the following features:
  - Load balancing
  - Performance and scalability of IT-resources
  - Availability and reliability of IT resources

- In Dynamic Failure Detection and Recovery Architecture, The resilient watchdog module performs the following five core functions:
  - Monitoring
  - Identifying an event
  - Executing the reactive routine/s
  - Reporting

- Several factors affect the Cloud reliability
  -

- Network Dependence
- Safety-Critical Processing

- Question : Write some threats in disaster recovery of cloud computing

  - Power failure or disruption
  - Computer Viruses
  - Fire, Flood & Disgruntled Employees
  - Lost Equipment & Desktop Failure
  - Server failure & Network Failure

- Dynamic Failure Detection and Recovery Architecture allows the implementation of an automated recovery policy consisting of predefined steps and may involve actions such as:

  - Running a script Sending a message
  - Restarting services

- Following are the steps in Bare-Metal Provisioning Architecture:

  - Consumer connects to central software through self-service portal.
  - The available servers are shown.
  - The consumer choses the server and the OS to be installed.
  - The resource management system is used to install the required chosen OS.
  - The consumer starts using the provisioned server.

- Following are the steps in Rapid Provisioning Architecture:

  - A consumer chooses a VM package through self-service portal and submits the provisioning request.
  - The centralized provisioning module selects an available VM and initiates it through a suitable template.
  - Upon initiation, the baseline/s templates are applied.
  - The VM is ready to use now.

- **[Cloud Consumer's Perspective]** The administrative rights of the IaaS consumer includes, controlling of:

  - Life cycle of VM (powering-On/Off and restarting)
  - Cloud storage attachment
  - Basic software installations (OS and pre-installed software)

  -

- Passwords and credentials
- Costs

- **[Cloud Consumer's Perspective]** Typically, a PaaS consumer receives the following:
  - Cloud emulation environment
  - Class libraries
  - Software libraries
  - Frameworks
  - Databases

- **[Cloud Consumer's Perspective]** The administrative rights of the PaaS consumer includes, controlling of:
  - Cloud storage device selection
  - Choosing the tools in case of ready-made environment
  - Deployment of automated scaling listener, load balancer and replication etc.
  - Login management of service/s developed/deployed using PaaS instance IT-resource usage cost

- **[Cloud Consumer's Perspective]** A few runtime configurations can be controlled by SaaS consumers. These include:
  - Security related configurations
  - SLA monitoring
  - Usage cost control

- Inter-Cloud providers are of two types:
  - With extra resources
  - With resource shortage

- **Business Cost Metrics:**
  - Upfront Costs
  - On-going Costs
  -

Cost management can take place across the lifecycle phases of Cloud services. These phases may include:

- Design & Development
- Deployment
- Service Contracting
- Provisioning & Decommissioning

The cost templates used by the providers depend upon:

- Market competition
- Overhead occurred during design, deployment and operations of the service
- Cost reduction considerations through increased sharing of IT resources

A pricing model for Cloud services can be composed of:

- Fixed and variable rates definitions
- Negotiations by consumers
- Cost customization possibilities
- Discount offerings
- Payment options

Two common metrics related to service measuring resiliency are as follows:

- Mean-Time System Recovery (MTSR) Metric
  Mean-Time to Switchover (MTSO) Metric

-

The key terminologies of privacy are:

- Data subject
- Data processor
- Data controller

Trust has two types:

- **Hard trust**
- Soft trust

- Dynamic trust (short term)
- Persistent trust (long term)

Five primary functions of Cryptography:

- Privacy
- Integrity
- Authentication
- Non-repudiation (A mechanism to prove the originality of the sender)
- Exchange of crypto keys which are the strings of bits used to change the format of the data

Remedies for Denial of Service (DoS) attacks:

- Contact ISP to clarify the reason of downgraded network performance
- ISP can help in throttling malicious traffic
- Using DoS detection tools

Some prominent security protocols for wireless security are:

- Wired Equivalent Privacy (WEP)
- Wi-Fi Protected Access (WPA)
- Wi-Fi Protected Access 2 (WPA2)

Encryption helps in countering:

- Insufficient authorization

https://chat.whatsapp.com/GecIrau2Nit0D4F5DjDiGM

- Traffic eavesdropping
- Malicious intermediary
- Overlapping trust boundaries

There are two basic types of encryption:

- Symmetric Encryption
- Asymmetric Encryption

Identity and Access Management (IAM) Consist of four main components:

- Authentication (verifies user identities)
- Authorization (grants access to specific resources based on user roles and permissions)
- User management (creates, updates, and deletes accounts as needed)
- Credential management

Several factors affect the Cloud reliability:

- Network Dependence
- Safety-Critical Processing

The system requirements are of two types:

- Functional (What the cloud service should do)
- Non functional (How well the cloud service should perform)

Cloud Solution Design Metrics:

- Accessibility
- Audit
- Availability
- Maintainability
- Backup

In Cloud Resource Scheduling, There are two main steps:

- Resource provisioning
- Resource scheduling

https://chat.whatsapp.com/GecIrau2Nit0D4F5DjDiGM

Effective resource scheduling reduces:

- Energy consumption
- Execution time
- Execution cost

Cloud Resource Scheduling Fulfills following QoS requirements:

- Security
- Availability
- Reliability
- Scalability

Types of computational offloading:

- Static
- Dynamic

Generic characteristics of big data:

- Data volume is huge
- The data is generated, captured and processed at high speed.
- Data may not be categorized into regular relational databases

The characteristics of big data can be represented by 4vs:

- Value
- Volume
- Velocity
- Variety

popular models for big data:

- Distributed Map Reduce model (popularized by Hadoop)
- NoSQL model (used for non-relational, non-tabular storage)
- SQL RDBMS model (for relational tabular storage of structured data)

https://chat.whatsapp.com/GecIrau2Nit0D4F5DjDiGM

Software Defined Networking (SDN) has four characteristics:

- Standardized APIs
- Centralization of control plane
- Programmable control plane
- Separation of control and data planes

Online gaming consists of four basic building blocks:

- Networking module
- Input module
- Rendering module
- Game logic module

Advantages of Mobile Cloud Computing:

- Extending battery lifetime
- Improving data storage capacity and processing power
- Improving reliability
- Additionally

https://chat.whatsapp.com/GecIrau2Nit0D4F5DjDiGM